

Predicting Changes in Rent Prices in Barcelona

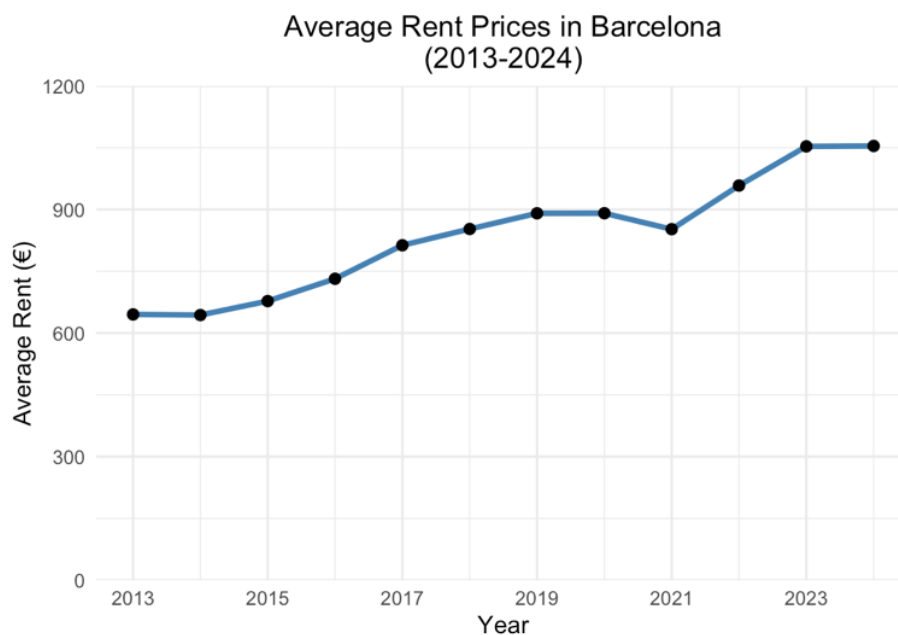
INTRODUCTION: A Prediction Problem

I predict changes in rental price in Barcelona by neighborhood using socioeconomic data on the population. Barcelona is well-known in the open data community for its rich and freely available data sets, making it an ideal case study for building a rich data panel.

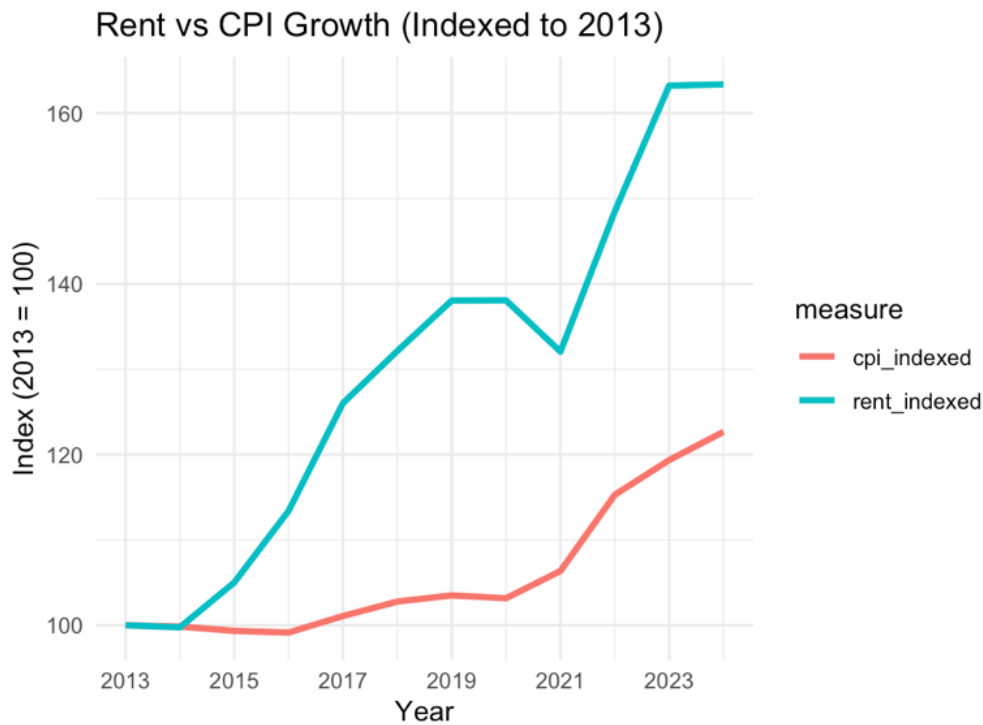
This is a relevant project because rent prices have a major impact on the standard of living of the population, and Barcelona is an often-cited example of European cities undergoing cost-of-living crises. A model for predicting rental costs could be beneficial from a social planning perspective, since benefits and subsidies can be indexed on effective housing cost predictions. More ambitiously, being able to predict rental costs with statistical significance could also reveal underlying correlations in a longer-term perspective. Such tools can be used to answer the question of how price increases can be moderated or kept under control, although this last objective goes beyond the scope of this project and the predictive model that I construct.

DATA DESCRIPTION: $[n = 840, p = 31]$

The observations for my data set are the rent prices for Barcelona's 73 districts, which are available from the years 2013 to 2024. This rich data set yields 864 observations. To ensure a statistically reliable set of observations, I dropped the neighborhoods that were missing more than 2 out of 12 years of data, which brought my final total number of observations to 840. Each row in my data panel represents a district of Barcelona in a given year between 2013 and 2024. The question then became, what kind of data should I seek to collect in order to predict rent increases? The context of rent increases helped to answer this problem.



(Figure 1)



(Figure 2)

Figure 1 shows the trend in average rent prices across Barcelona from 2013 to 2024, revealing a significant increase from approximately €640 in 2013 to €1,150 by 2024. **Figure 2** compares rent growth to the country-wide CPI (Consumer Price Index) growth, both indexed to 2013 levels, demonstrating that rents have risen substantially more than general prices have. These patterns motivated my selection of both macroeconomic variables (like CPI) and neighborhood-specific socioeconomic factors (population composition, income levels, migration patterns) as potential predictors of rent variation, but the gap between CPI and rent also suggested that there may be some unexpected or random relationships at play in the data. This encouraged me to include variables without obvious relationships to rent (e.g. deaths per neighborhood per year) and to use the Random Forest model as one of my methodologies.

To build my data panel, I used exclusively time-variant data that fit the time frame of my outcome observations. Not all data, though, was available for the full period 2013-2024, and some of my variables only cover select years during this period. The majority of my data was gathered in individual data sets (often with each year stored in a separate file) from the Barcelona municipality's open data website, Open Data BCN. Below is the full list of websites from which data was gathered together with the corresponding category of data:

1. **Open Data BCN:** <https://opendata-ajuntament.barcelona.cat>
For population data including population, births, deaths, education level, population density, nationality, unemployment, etc.
2. **Barcelona Municipal Statistics Office:** <https://portaldades.ajuntament.barcelona.cat>
For economic data including average taxable income.
3. **Generalitat / INCASÒL:** <https://habitatge.gencat.cat>
For outcome variable, rents in Barcelona.
4. **OHB – Observatori Metropolità de l'Habitatge:** <https://www.ohb.cat>
For housing sales.

DATA CLEANING AND PREPARATION

Data gathering and cleaning was the most time-consuming part of the project. Building my full panel required loading and merging roughly 150–200 files; while most were well-formatted, the remaining datasets caused major bottlenecks. Several came as messy Excel spreadsheets with inconsistent structures, and others split observations across separate yearly files, requiring careful concatenation. Despite extensive effort, some data quality issues may still remain, as fully verifying every variable across all neighborhoods and years was beyond the project's time constraints.

One illustrative challenge involved inconsistent neighborhood naming conventions across datasets. The neighborhood "el Poble-sec" appeared variously as "el Poble Sec," "el Poble Sec - AEI Parc Montjuïc," and "Poble Sec" across different data sources. These spelling variations prevented automatic matching during join operations, causing key observations to be dropped from the merged panel. To address this, I developed a standardization function that identified and corrected common naming patterns using string detection, ensuring that all variations mapped to a single consistent name. This function was applied to every dataset before merging, though I acknowledge that some less obvious naming inconsistencies may persist.

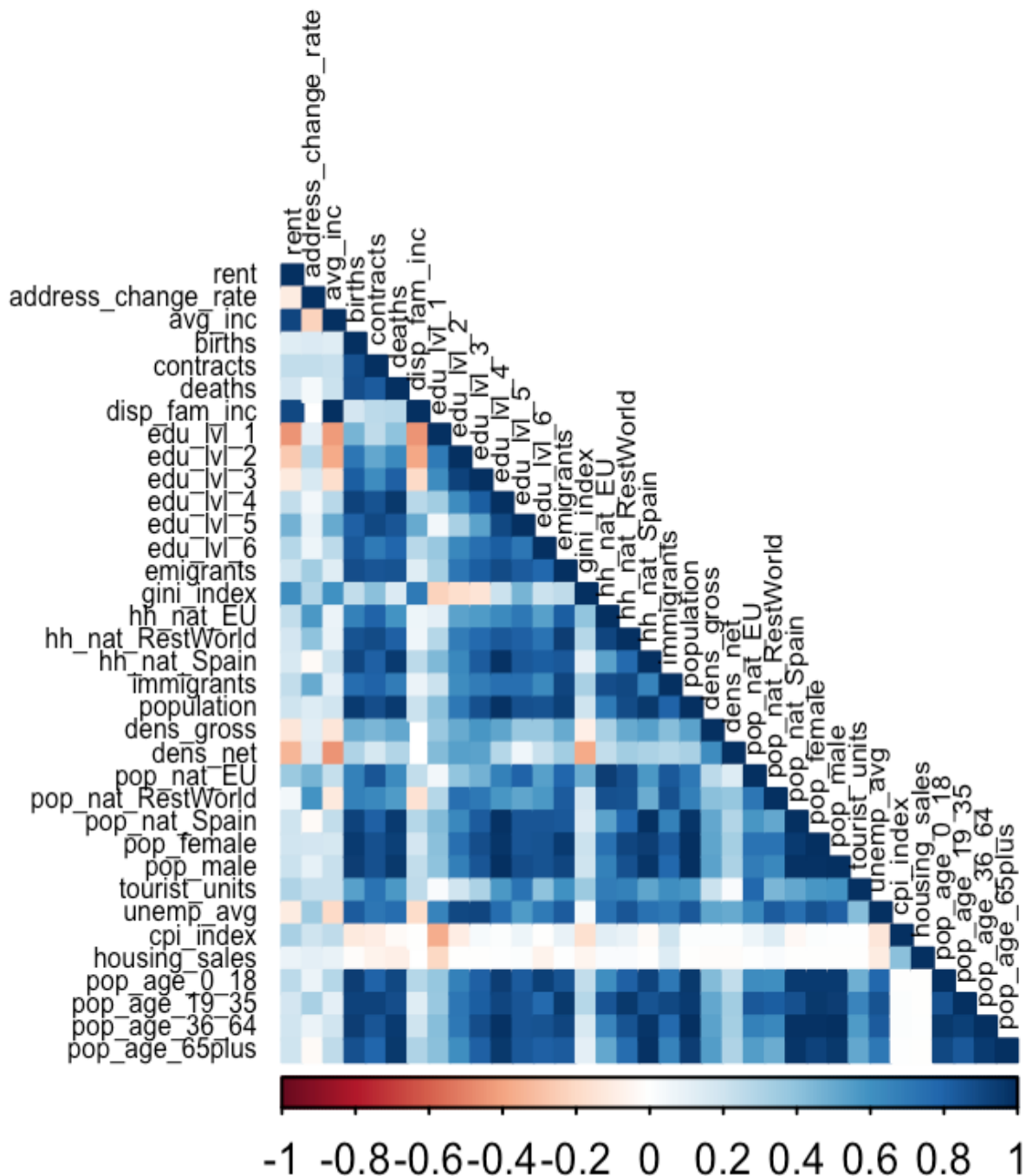
Multicollinearity

While multicollinearity is less problematic for predictive models than for inference, I removed variables with extremely high correlations ($r > 0.95$) to improve model interpretability. I dropped disposable family income ($r = 0.996$ with average income, and only available for 2022) and Spanish household nationality ($r = 0.996$ with Spanish population nationality), as these were redundant measures. I also created a sex ratio variable (`pct_female`) and removed the separate male and female population counts, which were near-perfectly correlated with total population ($r > 0.99$). These adjustments reduced redundancy while retaining conceptually distinct predictors across demographic composition, economic indicators, and migration patterns.

Feature Engineering

Beyond addressing multicollinearity, I made two key feature engineering decisions. First, I aggregated the 22 age group variables into four broader categories (0-18, 19-35, 36-64, and 65+), which reduced dimensionality from 22 to 4 variables while preserving meaningful information about neighborhood age composition. Second, I applied a log transformation to the outcome variable (rent) to address right-skewness in the distribution and ensure positive predictions after back-transformation. I kept these transformations conservative, as methods like Random Forest can capture non-linear relationships without explicit feature engineering, while linear regression and LASSO benefit from these foundational adjustments.

Full Correlation Matrix (2015-2023)



The correlation matrix, containing data from 2015-2023 to eliminate years with missing data, shows that neighborhood characteristics in Barcelona tend to move together in predictable ways, with clear clusters around education, income, and population nationality. Only a few variables reached problematic correlation levels, supporting a feature set that is rich but not excessively collinear for downstream modeling.

METHODOLOGY & RESULTS

LINEAR REGRESSION

Justification: I begin with a linear regression as a baseline model because it provides interpretable coefficients showing how each predictor affects rent prices and serves as a benchmark for comparing more complex methods. I apply a log transformation to the outcome variable (rent) to address potential right-skewness in the rent distribution and to ensure predictions remain positive, as the exponential back-transformation guarantees non-negative predicted values. This transformation also allows coefficients to be interpreted as approximate percentage effects, though the primary prediction task remains estimating actual rent levels in euros.

Results

Linear Regression Performance on Training Set

Metric	Value
RMSE (€)	44.97
MAE (€)	29.83
R ²	0.96

RMSE and MAE in euros; R² represents variance explained

Linear Regression Performance on Test Set

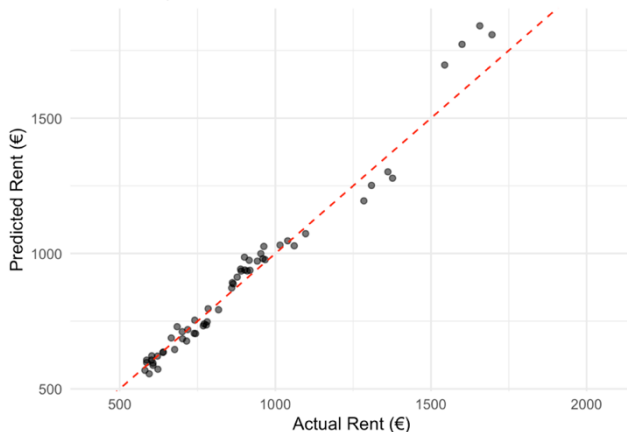
Metric	Value
RMSE (€)	55.83
MAE (€)	39.82
R ²	0.975

RMSE and MAE in euros; R² represents variance explained

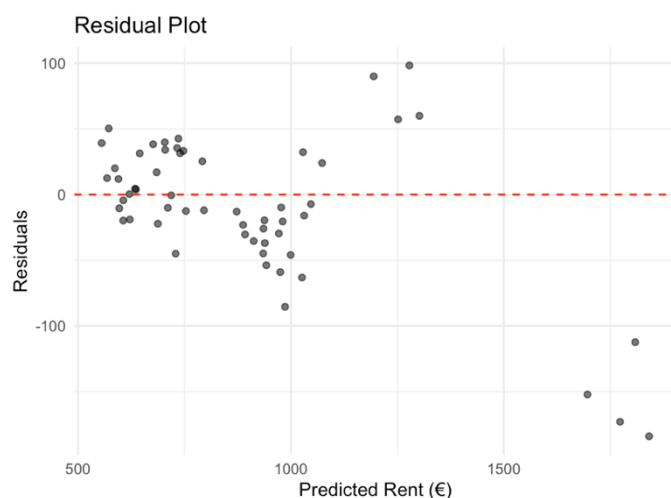
Table interpretation: The linear regression model demonstrates strong predictive performance on the test set, explaining 97.5% of variance in rent prices ($R^2 = 0.975$). With an average prediction error of €55.83 (RMSE) and typical error of €39.82 (MAE), the model accurately predicts neighborhood rent levels within approximately 5-8% of actual values. Performance on the training set is similarly strong ($R^2 = 0.96$, RMSE = €44.97), with the slightly better training metrics indicating minimal overfitting. These results establish a solid baseline for comparison with more complex methods like LASSO and Random Forest.

According to the linear regression coefficients, average income emerges as the dominant predictor of rent levels, with a €10,000 increase in neighborhood income associated with approximately 46% higher rents, while population density, nationality composition (EU, Rest of World, and Spanish populations), and neighborhood age structure also show significant predictive power.

Linear Regression: Predicted vs Actual Rent



The predicted versus actual rent plot shows strong agreement between model predictions and observed values, with points clustering tightly around the diagonal line. This indicates the linear regression model accurately captures rent price variation across neighborhoods and years.



The residual plot displays errors scattered randomly around zero with no systematic pattern, confirming the model's assumptions are met. A slight increase in variance at higher predicted rents suggests the model is marginally less precise for expensive neighborhoods, though overall fit remains strong.

Lasso – the wrong model !

Justification: While the linear regression performs well, it includes all available predictors without major feature selection, potentially incorporating irrelevant variables and risking overfitting despite strong test performance. LASSO (Least Absolute Shrinkage and Selection Operator) regression applies L1 regularization to automatically shrink coefficients of less important variables to zero, performing both regularization and feature selection simultaneously. I decided to use LASSO when I originally thought my data panel would have over 100 columns. The results below show why my project evolved in such a way as to make LASSO completely inadequate.

LASSO Performance

Metric	Validation	Test
RMSE (€)	157.57	135.2
MAE (€)	104.46	90.36
R ²	0.651	0.687
Variables Retained	26/26	—

Note: even when LASSO with a larger penalty parameter (lambda.1se rather than lambda.min) was applied, dropping 13 of 26 variables, the results were similar

LASSO underperformed compared to OLS (test R² of 0.69 versus 0.975), reflecting two key limitations. First, LASSO's complete-case requirement forced me to drop six variables with substantial missing data (tourist units, address change rate, average income, gini index, and both density measures), losing predictive information the linear model retained. Second, despite regularization, LASSO retained all 26 remaining predictors with heavily shrunk coefficients rather than performing feature selection, indicating no clearly irrelevant variables exist. The aggressive coefficient shrinkage reduced accuracy when most socioeconomic variables genuinely contribute to rent prediction, demonstrating that regularization hinders performance when predictors are broadly relevant with strong linear relationships.

Random Forest

Justification: Given the exceptionally strong performance of ordinary linear regression ($R^2 = 0.975$), I employ Random Forest as a robustness check and to test whether non-linear relationships or complex interactions might further improve predictions. RF can capture non-linearities and variable interactions without requiring explicit specification, making it ideal for exploring whether the linear model has missed more complex patterns in the data. Additionally, RF provides variable importance rankings without multicollinearity issues, offering complementary insights into which predictors most strongly drive rent variation.

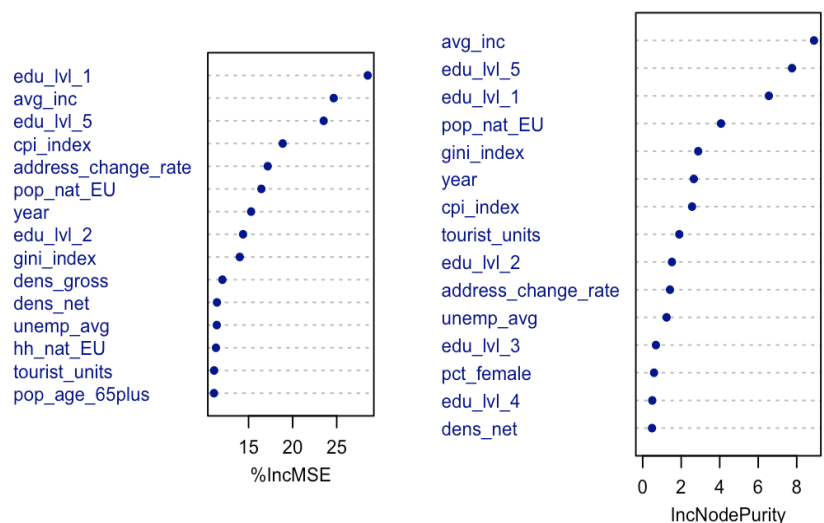
Random Forest Performance

Metric	Validation	Test
RMSE (€)	46.850	47.690
MAE (€)	36.140	36.490
R^2	0.959	0.937

Random Forest with the complete predictor set demonstrates exceptional performance (test $R^2 = 0.937$, RMSE = €47.69), approaching and in key metrics surpassing OLS. The lower RMSE and MAE relative to OLS (€47.69 vs €55.83, €36.49 vs €39.82) indicate that RF handles prediction errors more effectively, particularly for outliers, while the comparable R^2 values confirm both methods capture the underlying relationships equally well. This convergence validates that the identified socioeconomic predictors robustly explain Barcelona's rent variation across fundamentally different modeling approaches.

The variable importance plot displays the top 15 predictors ranked by their contribution to Random Forest model accuracy. The left panel (%IncMSE) shows how much prediction error increases when each variable is randomly permuted, with average income emerging as the dominant predictor, followed by education levels (edu_lvl_1, edu_lvl_5), CPI, and address change rate—removing these variables substantially degrades model performance. The right panel confirms that average income, education levels, and population nationality are the primary drivers of rent variation across Barcelona's neighborhoods.

Top 15 Most Important Variables



CONCLUSIONS

This analysis shows that Barcelona's neighborhood rent variation from 2013–2024 is driven by socioeconomic and demographic factors effectively modeled through multiple approaches. Ordinary least squares regression delivered outstanding predictive performance ($R^2 = 0.975$, RMSE = €55.83), explaining 97.5% of rent variance through linear combinations of income, education levels, temporal trends, population composition, and economic indicators.

LASSO underperformed ($R^2 = 0.69$) because its complete-case requirement excluded key variables including average income, and its regularization retained all remaining predictors with heavily shrunk coefficients. This suggests each socioeconomic factor contributes relevant information, making aggressive regularization counterproductive.

Random Forest with the complete predictor set closely matched OLS performance ($R^2 = 0.937$, RMSE = €47.69) and achieved lower absolute prediction errors (MAE = €36.49 vs €39.82). RF's variable importance rankings independently confirmed average income as the dominant predictor, followed by education levels, CPI, and population nationality.

The convergence of OLS and Random Forest validates these findings' robustness. While linear relationships dominate, Random Forest's competitive and in some metrics superior performance suggests subtle non-linearities may contribute. This consistency across fundamentally different methodologies strengthens confidence that average income, education, temporal trends, and demographic composition genuinely drive neighborhood-level rent variation, making either approach viable for forecasting Barcelona's housing market.